

Sardar Patel Institute of Technology, Mumbai

Department of Computer Engineering | Mini Project, Semester V | A.Y. 2026–27

Software Requirements Specification

MediScanX

*Online-Based Multimodal AI Diagnostic System for Thoracic and Cardiac Disease Detection***Project Team**

Name	UID	Responsibility
Vidhaan Shah	2024300212	Pipeline and database
Gaurang Satone	2024300205	frontend and UI
Jay Pradhan	2024300184	Integration and planning

Project Guide: Prof. Anand Godbole**Course Link:** Python Programming for Data Science, Database Management, Software Engineering**Revision History**

Version	Date	Description	Author
0.1	Aug 2026	Initial problem definition and scope (Phase I)	Team
1.0	30 Sep 2026	Complete SRS baseline for implementation	Team

Table of Contents

1. Introduction

- 1.1 Purpose
- 1.2 Scope
- 1.3 Definitions and Abbreviations
- 1.4 References

2. Overall Description

- 2.1 Product Perspective
- 2.2 User Classes
- 2.3 Operating Environment
- 2.4 Constraints
- 2.5 Assumptions and Dependencies

3. Modules and Functional Requirements

4. System Working and Design

- 4.1 End-to-End Workflow
- 4.2 Imaging Branch
- 4.3 ECG Branch
- 4.4 Multimodal Fusion
- 4.5 Explainability

5. Data Design

6. External Interface Requirements

- 6.1 User Interface
- 6.2 Software Interface (REST API)
- 6.3 Hardware Interface

7. Non-Functional Requirements

8. Verification and Validation

- 8.1 Model Evaluation
- 8.2 Software Testing

9. Risks, Limitations and Project Plan

- 9.1 Risks and Limitations
- 9.2 Project Plan

1. Introduction

1.1 Purpose

This document specifies the software requirements for MediScanX, a web-based decision-support system that analyses chest X-ray (and single-slice chest CT) images together with 12-lead ECG recordings to flag likely thoracic and cardiac abnormalities. It is written for the project guide and evaluators, and serves as the reference for design, implementation and testing by the team. It follows the structure of IEEE 830 / ISO/IEC/IEEE 29148, trimmed to what a semester project needs.

1.2 Scope

MediScanX lets a registered user create a patient case, upload a chest image and/or an ECG, and receive (a) probabilities for a fixed set of thoracic and cardiac conditions, (b) a visual explanation of what the model focused on, (c) a combined cardiac-concern summary when both modalities are present, and (d) a downloadable PDF report. All models are trained on public research datasets.

Out of scope: 3D CT volume analysis; real-time or streaming ECG; integration with hospital PACS/HIS; treatment or drug recommendations; use with real patient data outside a supervised demo; regulatory certification. MediScanX is an academic prototype and a second-opinion aid, never a substitute for a radiologist or cardiologist.

1.3 Definitions and Abbreviations

Term	Meaning
CXR / CT / ECG	Chest X-ray / Computed Tomography / Electrocardiogram
CNN	Convolutional Neural Network; 1D CNN is applied to ECG time series
Grad-CAM	Gradient-weighted Class Activation Mapping, an XAI method that highlights influential regions
XAI	Explainable Artificial Intelligence
DICOM / WFDB	Standard medical image format / PhysioNet waveform format (.hea + .dat)
AUROC	Area under the ROC curve, the main model-quality metric used here
PTB-XL	Public 12-lead ECG dataset (21,799 records) from PhysioNet
JWT / RBAC	JSON Web Token / Role-Based Access Control
MoSCoW	Priority scale used in this SRS: M = Must, S = Should, C = Could

1.4 References

- IEEE Std 830-1998 and ISO/IEC/IEEE 29148:2018, software requirements specification guidelines.
- Wang et al., ChestX-ray8/14, CVPR 2017; Rajpurkar et al., CheXNet, arXiv 2017 (imaging baseline).
- Wagner et al., PTB-XL, Scientific Data 2020; Strodthoff et al., Deep learning for ECG analysis: benchmarks on PTB-XL, IEEE JBHI 2021.
- Huang et al., DenseNet, CVPR 2017; Tan and Le, EfficientNet, ICML 2019; Selvaraju et al., Grad-CAM, ICCV 2017.

2. Overall Description

2.1 Product Perspective

MediScanX is a self-contained three-tier web application: a React/Next.js front end, a Python (FastAPI) back end that hosts preprocessing and model inference, and a PostgreSQL database with file storage for uploads and generated artefacts (Figure 1, Section 4). It does not depend on any external clinical system. Models are trained offline on free GPU tiers (Google Colab / Kaggle) and served on CPU; the final decision always rests with the clinician.

2.2 User Classes

User class	Description	Main actions
Clinician / Medical user	Primary user; doctor, radiology or cardiology trainee, or student evaluator	Create cases, upload studies, view results and heatmaps, add remarks, download reports
Administrator	Team member or faculty who manages the deployment	Manage users, view audit logs, register and activate model versions

2.3 Operating Environment

- **Client:** latest two versions of Chrome, Edge or Firefox on desktop or tablet.
- **Server:** Ubuntu 22.04 (or equivalent cloud instance), 4 vCPU, 8 GB RAM, Python 3.10+, PostgreSQL 15; inference runs on CPU, GPU is used only for training.
- **Libraries:** PyTorch or TensorFlow/Keras, OpenCV, SciPy, NumPy, pydicom, wfdb, SQLAlchemy, ReportLab (PDF).

2.4 Constraints

- Only public, de-identified datasets are used for training; no real patient data is collected by the team.
- Compute is limited to free GPU sessions, so model sizes and training schedules are kept modest (transfer learning, not training from scratch).
- The project must show a working end-to-end demo at the End Semester Evaluation in early November 2026.
- Only open-source tools are used; the system must be served over HTTPS.

2.5 Assumptions and Dependencies

- Uploaded X-rays are frontal (PA/AP) views; ECGs are standard 12-lead, 10-second recordings.
- NIH ChestX-ray14 and PTB-XL remain publicly downloadable; ImageNet-pretrained weights are available.
- No public dataset gives same-patient CXR + ECG pairs without credentialed access (MIMIC-CXR and MIMIC-IV-ECG need PhysioNet credentialing). The design therefore does not depend on paired data (see Section 4.4).

3. Modules and Functional Requirements

The system is divided into ten modules. Table 1 gives the overview; Table 2 lists the numbered requirements that will be traced to design and test cases.

M1	User and Access Management	Registration, login, roles	SE / DBMS
M2	Patient and Case Management	Anonymised patients, cases, history	DBMS
M3	Upload and Validation	Accept and check images and ECG files	SE
M4	Preprocessing	Standardise images and ECG signals	Python for DS
M5	Imaging Analysis	CXR (and single CT slice) classification	Python for DS
M6	ECG Analysis	ECG classification and basic metrics	Python for DS
M7	Multimodal Fusion	Combine both branches into a case summary	Python for DS
M8	Explainability	Grad-CAM heatmaps and ECG saliency	Python for DS
M9	Results and Reporting	Results page, PDF report, clinician remarks	SE
M10	Administration and Audit	User control, audit log, model registry	DBMS / SE

Table 1: Module overview

ID	Requirement (the system shall...)	Pri.
M1 User and Access Management		
FR-1.1	allow registration and login with email and password; passwords stored as bcrypt hashes	M

ID	Requirement (the system shall...)	Pri.
FR-1.2	issue JWT access tokens that expire after 60 minutes	M
FR-1.3	enforce role-based access for the User and Admin roles on every endpoint	M
FR-1.4	let a logged-in user change their password	S
M2 Patient and Case Management		
FR-2.1	create a patient record using an anonymised ID, age and sex only (no name or contact details)	M
FR-2.2	create a case linked to a patient, holding one or more uploaded studies and their results	M
FR-2.3	list, search and filter cases by case ID, date and status	M
FR-2.4	show a patient's previous cases and predictions	S
FR-2.5	delete a case together with its stored files and results	M
M3 Upload and Validation		
FR-3.1	accept chest images as JPEG/PNG (M) and DICOM (S), up to 20 MB each	M
FR-3.2	accept ECG as CSV with 12 lead columns (M) and WFDB .hea/.dat pairs (S)	M
FR-3.3	validate type, size, image dimensions and ECG lead count, and return a clear error message on failure	M
FR-3.4	allow analysis with only one modality present	M
FR-3.5	remove patient metadata from DICOM headers before storage	S
M4 Preprocessing		
FR-4.1	convert images to 3-channel 224×224 tensors normalised with ImageNet statistics; optional CLAHE contrast step	M
FR-4.2	resample ECG to 100 Hz, apply a 0.5–40 Hz Butterworth band-pass filter, normalise per lead and pad/trim to 10 s (12×1000)	M
FR-4.3	flag near-blank or very low-contrast images and flat or highly noisy ECG leads as poor quality	S
M5 Imaging Analysis		
FR-5.1	output a probability for each of 14 ChestX-ray14 labels (multi-label, sigmoid outputs)	M
FR-5.2	mark findings above per-label thresholds chosen on the validation set	M
FR-5.3	record the model version and inference time with each prediction	M
FR-5.4	accept a single axial CT slice and give a normal/abnormal indication using a separately fine-tuned lightweight model	C
M6 ECG Analysis		
FR-6.1	output probabilities for the five PTB-XL superclasses: NORM, MI, STTC, CD, HYP	M
FR-6.2	render the uploaded ECG as a 12-lead plot in the results page	M
FR-6.3	estimate heart rate using R-peak detection (scipy.signal.find_peaks)	S
FR-6.4	record the model version and inference time with each prediction	M
M7 Multimodal Fusion		
FR-7.1	compute a case-level cardiac-concern score by decision-level weighted fusion of both branches (Section 4.4)	M
FR-7.2	fall back to the single available modality and state which modalities were used	M
FR-7.3	flag disagreement between modalities (e.g. cardiomegaly on CXR but normal ECG) for clinician attention	M
FR-7.4	support feature-level fusion (embedding concatenation + MLP) if paired data is obtained, and compare it in an ablation	C
M8 Explainability		
FR-8.1	overlay a Grad-CAM heatmap on the X-ray for the top predicted label	M
FR-8.2	let the user choose which label to explain and adjust overlay opacity	S

ID	Requirement (the system shall...)	Pri.
FR-8.3	highlight the ECG segments that contributed most (1D Grad-CAM/saliency) on the lead plot	S
FR-8.4	state on every explanation that highlighted regions show model attention, not proven cause	M
M9 Results and Reporting		
FR-9.1	show probabilities, heatmap, ECG plot, fusion summary and quality warnings on one results page	M
FR-9.2	generate a downloadable PDF report with case ID, date, inputs, results, heatmap, model versions and disclaimer	M
FR-9.3	let the clinician add remarks and mark each AI finding as agreed or disagreed	S
M10 Administration and Audit		
FR-10.1	let an admin activate or deactivate user accounts	M
FR-10.2	log login, upload, prediction and deletion events with user ID and timestamp	M
FR-10.3	keep a registry of model versions and let an admin choose the active one	S
FR-10.4	show basic usage statistics (cases per day, average inference time)	C

Table 2: Functional requirements (M = Must, S = Should, C = Could)

4. System Working and Design

4.1 End-to-End Workflow

1. The user logs in, creates or selects a patient, and opens a new case.
2. The user uploads a chest image and/or an ECG file. The API validates format, size and content (FR-3.x).
3. The preprocessing service prepares the image (resize, normalise) and the ECG (resample, filter, normalise).
4. The imaging branch and ECG branch each run inference and return class probabilities.
5. The fusion module combines the available outputs into a cardiac-concern score and checks for disagreement.
6. The explainability module generates a Grad-CAM heatmap (and ECG saliency) for the top findings.
7. Predictions, heatmaps, model versions and timings are saved to PostgreSQL and file storage.
8. The results page is shown; the user can add remarks and download the PDF report.

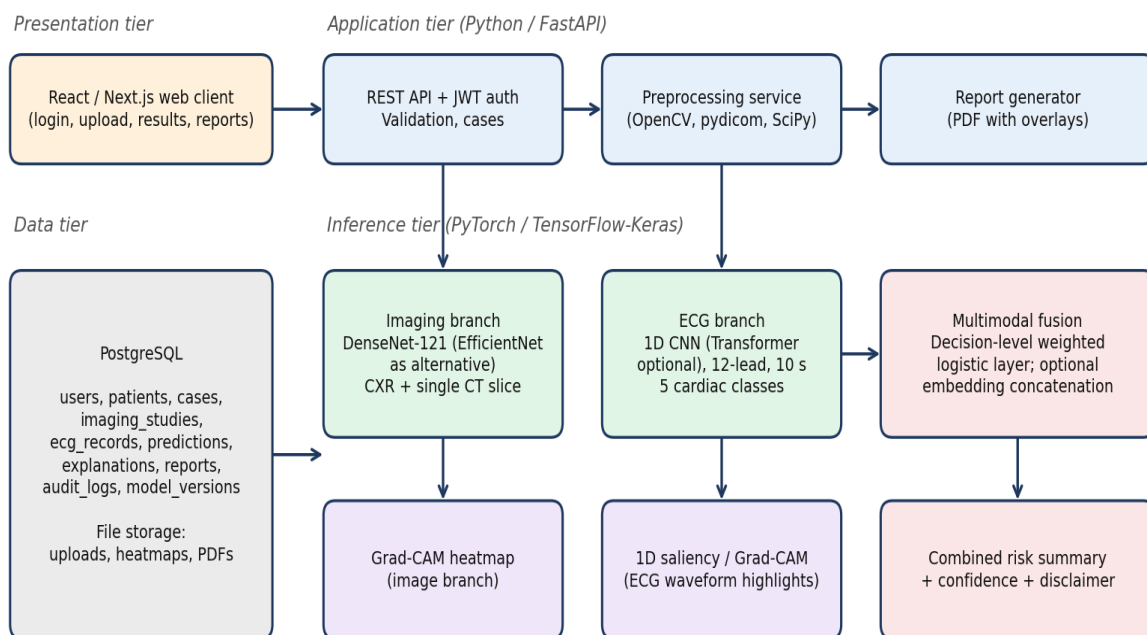


Figure 1: Logical architecture and inference flow of MediScanX

4.2 Imaging Branch

Data: NIH ChestX-ray14 (about 112,000 frontal X-rays, 14 labels), using the official patient-wise train/validation and test lists to avoid patient leakage. **Model:** DenseNet-121 initialised from ImageNet weights (the CheXNet baseline); EfficientNet-B0 is the fallback if training time is short. The final layer is replaced by 14 sigmoid outputs. **Training:** input 224×224, weighted binary cross-entropy to handle class imbalance, Adam (lr 1e-4), batch size 32, early stopping on validation AUROC. Augmentation is limited to small rotations, crops and brightness changes; horizontal flips are avoided because they alter cardiac position. **Thresholds:** per-label thresholds are picked on the validation set using Youden’s J statistic.

4.3 ECG Branch

Data: PTB-XL at 100 Hz, five diagnostic superclasses (NORM, MI, STTC, CD, HYP), with the recommended stratified folds (1–8 train, 9 validation, 10 test). **Model:** a compact 1D residual CNN (four convolutional blocks with batch norm and ReLU, global average pooling, five sigmoid outputs) taking a 12×1000 input; a small Transformer encoder is an optional experiment. **Training:** binary cross-entropy, Adam, early stopping on validation macro AUROC. Published benchmarks on PTB-XL report macro AUROC around 0.92–0.93 for strong models, so a lighter model is expected to land somewhat below that.

4.4 Multimodal Fusion

The two branches are trained independently on public data, so the primary fusion is **decision-level** and needs no paired dataset. For the cardiac concern, the imaging evidence is $p_{img} = \max(\text{Cardiomegaly, Edema, Effusion})$ and the ECG evidence is $p_{ecg} = 1 - P(\text{NORM})$. The case score is:

$$S = (w_{img} \cdot p_{img} + w_{ecg} \cdot p_{ecg}) / (w_{img} + w_{ecg}), \text{ where each weight equals that branch's validation AUROC on the relevant labels.}$$

If only one modality is present, its score is used alone. If $|p_{img} - p_{ecg}|$ exceeds 0.5, the case is flagged as “modalities disagree”. Scores are shown as Low / Moderate / High bands and described as a screening indicator, not a diagnosis. **Stretch goal:** if PhysioNet credentialed access is granted in time, penultimate-layer embeddings (1024-d image, ECG vector) are concatenated and passed to a small MLP trained on linked MIMIC records, and an ablation compares image-only, ECG-only, decision-level and feature-level fusion.

4.5 Explainability

Grad-CAM takes the gradient of the chosen class score with respect to the last convolutional feature maps A^k , averages it over spatial positions to get weights α^k , and forms the map $\text{ReLU}(\sum \alpha^k A^k)$. This is upsampled to the input size, normalised and overlaid on the X-ray as a colour heatmap. The same idea applied to the last 1D convolution of the ECG model gives a time-axis importance curve that is drawn over the lead plots. Heatmaps are a plausibility check only. Their localisation is qualitatively compared with the roughly 1,000 bounding-box annotations that ship with ChestX-ray14.

5. Data Design

The PostgreSQL schema is designed to third normal form. Images, ECG files, heatmaps and PDFs live in file storage outside the web root; the database stores their paths and hashes.

Table	Key attributes	Relationships
users	user_id (PK), email (unique), password_hash, role, is_active, created_at	1 user : many cases
patients	patient_id (PK), anon_code (unique), age, sex	1 patient : many cases
cases	case_id (PK), patient_id (FK), created_by (FK), status, created_at	1 case : many studies, predictions
imaging_studies	study_id (PK), case_id (FK), modality (CXR/CT), file_path, sha256, quality_flag	belongs to case
ecg_records	ecg_id (PK), case_id (FK), file_path, sampling_rate, quality_flag	belongs to case

Table	Key attributes	Relationships
predictions	pred_id (PK), case_id (FK), source (image/ecg/fusion), label, probability, model_id (FK), inference_ms	belongs to case, model
explanations	expl_id (PK), pred_id (FK), type (gradcam/saliency), file_path	belongs to prediction
reports	report_id (PK), case_id (FK), file_path, remarks, generated_at	belongs to case
model_versions	model_id (PK), name, version, dataset, val_auroc, is_active	referenced by predictions
audit_logs	log_id (PK), user_id (FK), action, target_id, timestamp	belongs to user

Table 3: Core database tables

6. External Interface Requirements

6.1 User Interface

Six screens: Login/Register, Dashboard (recent cases), New Case/Upload, Results (probability bars, heatmap with opacity slider, ECG plot, fusion summary, disclaimer banner), Case History, and Admin Panel. The layout is responsive for desktop and tablet, and a first-time user should reach a result within five clicks after login.

6.2 Software Interface (REST API)

Method	Endpoint	Purpose
POST	/auth/register, /auth/login	Create account, obtain JWT
POST / GET	/patients, /patients/{id}	Create and fetch patients
POST / GET	/cases, /cases/{id}	Create case, list and fetch case with results
POST	/cases/{id}/upload	Upload image and/or ECG (multipart)
POST	/cases/{id}/analyze	Run preprocessing, inference, fusion and XAI
GET	/cases/{id}/report	Download the PDF report
DELETE	/cases/{id}	Delete case and files
GET / PATCH	/admin/users, /admin/models, /admin/audit	Admin operations

Table 4: Main API endpoints

6.3 Hardware Interface

No special hardware is required by the end user. The server needs the resources in Section 2.3; training uses a Colab/Kaggle GPU.

7. Non-Functional Requirements

Category	Requirement
Performance	End-to-end analysis (one CXR + one ECG, including Grad-CAM) completes within short time stamp on the reference server; pages load fast; several concurrent users are supported (demo scale).
Model quality (targets)	Imaging mean AUROC ≥ 0.80 across the 14 labels; ECG macro AUROC ≥ 0.87 on the PTB-XL test fold. These are goals; actual figures are reported honestly with confidence intervals.
Security	HTTPS only; bcrypt password hashing; JWT with expiry; RBAC on all routes; ORM parameterised queries; file type and size checks; uploads stored outside the web root; no direct patient identifiers stored.
Reliability	Failed inference or invalid input returns a readable error without crashing the service; database exported before every demo.
Usability	Clear labels, plain-language result descriptions and a visible disclaimer on all result pages and reports.
Maintainability	Modular code (one package per module), PEP 8, Git feature branches with pull requests, README with setup steps, pinned requirements.

Category	Requirement
Portability	Runs on any Linux or Windows machine using requirements.txt and an .env file; Docker packaging is optional.
Ethical / legal	Public de-identified data only; users can delete their cases; outputs are labelled as decision support, in the spirit of India's DPDP Act 2023 data-minimisation principle.

Table 5: Non-functional requirements

8. Verification and Validation

8.1 Model Evaluation

- Per-label and macro AUROC, plus sensitivity, specificity and F1 at the chosen thresholds, on held-out test sets (ChestX-ray14 test list; PTB-XL fold 10).
- Ablation: image-only vs ECG-only vs fusion on the cardiac-concern task, on whatever linked data is available.
- Grad-CAM sanity review on a sample of bounding-box-annotated images, reported as a qualitative check.

8.2 Software Testing

- **Unit tests (pytest):** preprocessing functions, fusion formula, threshold logic, file validators.
- **API tests:** each endpoint for valid input, invalid input and unauthorised access.
- **Manual UI tests:** a checklist covering upload, results, report download and admin actions.

ID	Acceptance criterion	Covers
AC-1	A user can register, log in, create a case, upload a CXR and ECG, and reach a results page	FR-1.x, 2.x, 3.x
AC-2	Invalid or oversized files are rejected with a clear message	FR-3.3
AC-3	Both branches return probabilities; fusion score and disagreement flag appear	FR-5.1, 6.1, 7.x
AC-4	A Grad-CAM overlay is shown for the top label with the disclaimer	FR-8.1, 8.4
AC-5	The PDF report downloads with all required sections	FR-9.2
AC-6	A non-admin user is blocked from admin routes; events appear in the audit log	FR-1.3, 10.2

Table 6: Acceptance criteria

9. Risks, Limitations and Project Plan

9.1 Risks and Limitations

Risk / limitation	Mitigation
ChestX-ray14 labels are text-mined and noisy, which caps achievable accuracy	Report AUROC (not accuracy), state the limitation, compare against CheXNet-level results
No paired CXR + ECG data openly available	Decision-level fusion as the baseline; feature-level fusion only as a stretch goal
Models may not generalise to other hospitals' images or ECG machines	State the limitation; use augmentation; do not claim clinical validity
Limited GPU time on free tiers	Transfer learning, smaller backbones, save checkpoints each epoch
Schedule pressure before the End Semester Evaluation	Work is ordered by priority: all Must items first, Should next, Could only if time remains
Over-reliance on AI output	Disclaimer everywhere, clinician remarks field, confidence bands and disagreement flag

Table 7: Key risks

9.2 Project Plan

Phase	Period	Planned deliverables
Phase I	Mid Aug – Sep 2026	Problem definition, literature review, SRS, Git repository, PostgreSQL schema, dataset download and exploration
Phase II	Sep – mid Oct 2026	Preprocessing pipelines, imaging and ECG model training and evaluation, backend API with authentication and case management
Phase III	Mid Oct – end Oct 2026	Fusion and Grad-CAM modules, frontend integration, PDF report, testing, deployment
End Semester Evaluation	Early Nov 2026	End-to-end demo, final report, evaluation results and presentation

Table 8: Phase-wise plan (dates to be aligned with the department schedule)

Mapping to subjects: Python Programming for Data Science (Modules M4–M8: data handling, model training, evaluation, visualisation); Database Management (M2, M10, Section 5: schema design, normalisation, queries); Software Engineering (this SRS, modular design, version control, testing, Section 8).